

Phishing Website Detection – Business Problem and Dataset Insights

1. Business Problem Summary

Phishing websites are fraudulent sites that impersonate trusted organizations with the goal of stealing sensitive user information such as login credentials, banking details, or personal data. These attacks represent one of the fastest-growing cybercrime threats, impacting individuals, enterprises, and governments worldwide.

The scope of the problem extends beyond financial losses: successful phishing attacks erode customer trust, damage organizational reputation, and can even compromise national security. The importance of early detection is therefore critical to safeguarding digital ecosystems.

The business objective is to design accurate, scalable, and adaptive detection systems capable of identifying phishing websites—even newly emerging ones—using advanced analytics, machine learning (ML), and deep learning (DL) approaches. Such systems must minimize false positives while maintaining robustness against attackers' evolving tactics.

2. Insights from Phishing Literature

1.1 Common Characteristics of Phishing Websites

- URL-Level Traits: Suspiciously long URLs, presence of excessive special characters (@, -, //), multiple subdomains, unusual extensions, and high entropy values.
- Domain-Level Traits: Recently registered domains, short registration lengths, domains containing digits or hyphens, or domains mimicking popular brand names.
- Structural Features: Invisible iframes, abnormal HTML tags, high number of external redirects.
- Content Features: Missing logos/images, fake login forms, urgency-triggering messages.
- Delivery Mechanism: Spread through spam emails, malicious ads, or social engineering strategies.

2.2 Detection Challenges

- Evasion Tactics: Phishers continuously adapt, registering new domains and altering content to bypass blacklists.
- Feature Dependence: Traditional systems rely on manually engineered features, which may not generalize to novel phishing methods.
- Data Challenges: Imbalanced datasets, rapidly changing phishing techniques (concept drift), and risk of false positives.

2.3 Potential Solutions

- Machine Learning Approaches: Feature-based ML models (Random Forest, XGBoost), ensemble methods.
- Deep Learning Approaches: CNNs, RNNs, and transformers on raw URL and HTML data.
- Hybrid Models: Combine URL, domain, and content features for robust analysis.
- Data-Centric Solutions: Large labeled datasets, synthetic data generation for imbalance correction.

3. Phishing Website Feature Summary

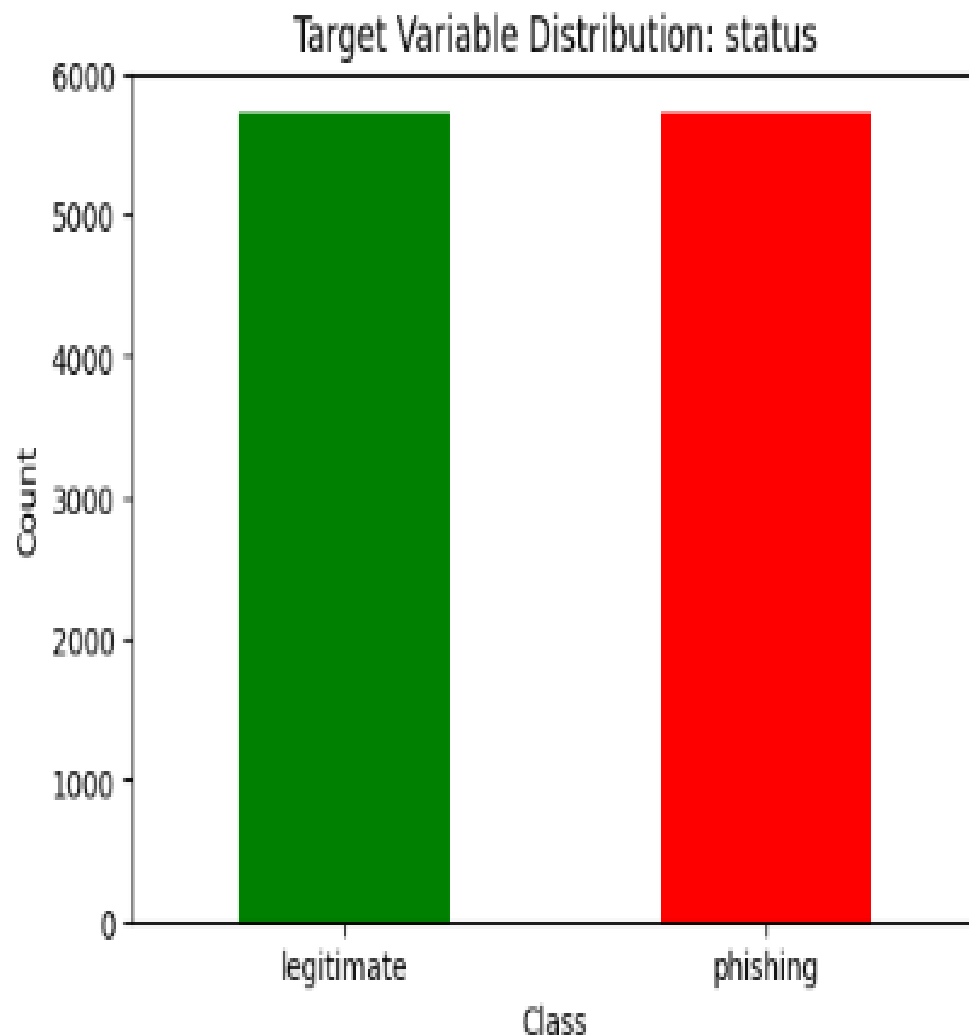
A wide range of features are used for phishing detection. These can be grouped as follows:

- A) URL Features: Length, number of digits, special characters, entropy, misplaced TLDs.
- B) Domain Features: Domain age, registration length, presence of hyphens/digits, brand similarity.
- C) Subdomain Features: Count and abnormal structures.
- D) Structural Features: Iframes, external/internal link ratios, suspicious HTML tags.
- E) Content Features: Lack of images, fake forms, suspicious words.
- F) Popularity Features: Page rank, Google index, web traffic.

4. Dataset Overview

The dataset used in this analysis contains 11,430 rows and 89 features, balanced between legitimate (5,715) and phishing (5,715) websites. The features capture URL structure, domain characteristics, content properties, and popularity metrics.

```
=== Target Variable Distribution ===
status
legitimate    5715
phishing      5715
Name: count, dtype: int64
```



Feature Summary (with Relevance for Phishing Detection)

Below is a structured summary of each feature:

- url (object) → The actual website URL (unique identifier). Useful for reference but not directly a predictive feature.
- length_url (int) → Length of the full URL. Long/random URLs often indicate phishing.
- length_hostname (int) → Length of the domain/host part. Longer hostnames can suggest suspicious domains.
- ip (int) → Whether the URL uses a raw IP instead of a domain (phishing indicator).
- nb_dots (int) → Number of . in the URL. More dots → possible obfuscation.
- nb_hyphens (int) → Count of - (hyphens). Attackers often use hyphens to mimic legitimate domains.
- nb_at (int) → Count of @. URLs with @ may redirect to another location.
- nb_qm (int) → Number of ?. Often used to append suspicious queries.
- nb_and (int) → Number of & operators in query string.
- nb_or (int) → Number of | operators (rare in normal URLs, suspicious if present).
- nb_eq (int) → Number of = signs. Excessive parameters may indicate phishing.
- nb_underscore (int) → Count of _ . Sometimes used to hide true domain.
- nb_tilde (int) → Count of ~. Rare, but may indicate non-standard links.
- nb_percent (int) → Number of % (URL encoding). Excessive encoding may signal obfuscation.
- nb_slash (int) → Number of /. Many redirects/subpaths → suspicious.
- nb_star (int) → Number of *. Rare in normal URLs, suspicious if found.
- nb_colon (int) → Number of :. Used in ports or alternate protocols.
- nb_comma (int) → Count of ,.
- nb_semicolumn (int) → Count of ;.
- nb_dollar (int) → Count of \$. Rare, may indicate tricks.
- nb_space (int) → Count of spaces in URL (encoded as %20).
- nb_www (int) → Count of "www" substrings. Multiple occurrences can be suspicious.
- nb_com (int) → Count of .com. If repeated, may signal fake domains.
- nb_dslash (int) → Number of // occurrences (beyond http://). Suspicious.
- http_in_path (int) → Presence of "http" inside path. Fake sites mimic protocols.
- https_token (int) → "https" appearing in non-standard places (fake security indicator).
- ratio_digits_url (float) → Ratio of digits in URL. High digit ratio = suspicious.
- ratio_digits_host (float) → Ratio of digits in hostname.
- punycode (int) → Whether internationalized domain names (IDN) are used. Some IDNs are malicious.
- port (int) → Use of uncommon ports (phishing risk).
- tld_in_path (int) → TLD string appearing in path (fake .com inside path).
- tld_in_subdomain (int) → TLD string appearing in subdomain (phishing trick).
- abnormal_subdomain (int) → Presence of unusually long or nested subdomains.
- nb_subdomains (int) → Count of subdomains. More → suspicious.
- prefix_suffix (int) → Use of - in domain prefix/suffix. Phishers often mimic brands like "paypal-secure-login".
- random_domain (int) → Randomly generated domain patterns.
- shortening_service (int) → Whether a URL shortener (bit.ly, tinyurl, etc.) is used. Common in phishing.
- path_extension (int) → Suspicious extensions in path.
- nb_redirection (int) → Number of redirections.
- nb_external_redirection (int) → Redirects to external domains.

- length_words_raw (int) → Total length of words in URL.
- char_repeat (int) → Repetition of same characters (like aaaa).
- shortest_words_raw (int) → Length of shortest word in URL.
- shortest_word_host (int) → Shortest word in hostname.
- shortest_word_path (int) → Shortest word in path.
- longest_words_raw (int) → Longest word length in URL.
- longest_word_host (int) → Longest word in hostname.
- longest_word_path (int) → Longest word in path.
- avg_words_raw (float) → Average word length in URL.
- avg_word_host (float) → Average word length in hostname.
- avg_word_path (float) → Average word length in path.
- phish_hints (int) → Count of suspicious keywords (like “login”, “secure”).
- domain_in_brand (int) → Domain matches known brand.
- brand_in_subdomain (int) → Brand name appears in subdomain (fake mimicry).
- brand_in_path (int) → Brand name appears in path.
- suspicious_tld (int) → Suspicious TLD like .xyz, .tk, etc.
- statistical_report (int) → Presence in known phishing statistical databases.
- nb_hyperlinks (int) → Number of hyperlinks in page.
- ratio_intHyperlinks (float) → Ratio of internal links.
- ratio_extHyperlinks (float) → Ratio of external links.
- ratio_nullHyperlinks (int) → Broken/empty links.
- nb_extCSS (int) → External CSS files linked.
- ratio_intRedirection (int) → Internal redirections.
- ratio_extRedirection (float) → External redirections.
- ratio_intErrors (int) → Internal errors.
- ratio_extErrors (float) → External errors.
- login_form (int) → Presence of login form.
- external_favicon (int) → Favicon hosted externally (phishing risk).
- links_in_tags (float) → Links within tags.
- submit_email (int) → Submitting credentials via email.
- ratio_intMedia (float) → Internal media links.
- ratio_extMedia (float) → External media links.
- sfh (int) → “Server Form Handler” (phishing risk).
- iframe (int) → Presence of iframe (used in phishing).
- popup_window (int) → Popup windows (phishing tactic).
- safe_anchor (float) → Ratio of safe anchors.
- onmouseover (int) → Onmouseover script trick.
- right_click (int) → Disabled right-click (phishers prevent inspection).
- empty_title (int) → Empty web page title.
- domain_in_title (int) → Domain appears in title.
- domain_with_copyright (int) → Domain name appears in copyright section.
- whois_registered_domain (int) → Domain has valid WHOIS registration.
- domain_registration_length (int) → How long the domain is registered (short → suspicious).
- domain_age (int) → Age of domain (new = suspicious).
- web_traffic (int) → Web traffic ranking.
- dns_record (int) → Valid DNS record exists.
- google_index (int) → Indexed by Google (missing index = suspicious).
- page_rank (int) → Page rank score.
- status (object) → Target variable: legitimate or phishing.

=== Column Details ===

	Column Name	Data Type	Non-Null Count	Null Count	Unique Values
0	url	object	11430	0	11429
1	length_url	int64	11430	0	324
2	length_hostname	int64	11430	0	83
3	ip	int64	11430	0	2
4	nb_dots	int64	11430	0	19
...	...	...	...	...	...
84	web_traffic	int64	11430	0	4744
85	dns_record	int64	11430	0	2
86	google_index	int64	11430	0	2
87	page_rank	int64	11430	0	11
88	status	object	11430	0	2

89 rows × 5 columns

5. Initial Dataset Observation

A) URL & String-Based Features

Includes length of URL/hostname, special character counts (dots, hyphens, at-symbols, etc.), misplaced tokens (HTTP/HTTPS), punycode, random domain, prefix/suffix, and shortening services.

B) Domain & Subdomain Features

Covers abnormal subdomains, number of subdomains, presence of brand names in domain/subdomain, domain copyright, and WHOIS information including domain age and registration length.

C) Content & HTML Features

Features include hyperlink counts, internal vs external ratios, CSS usage, redirection ratios, errors, presence of login forms, external favicons, submit-email functionality, iframes, popup windows, disabled right-click, and empty titles.

D) Statistical & Lexical Features

Word-level and character-level analysis: length of words, shortest/longest words, average word length, repeated characters, and phishing hints (keywords such as 'login', 'secure').

E) Popularity & Ranking Features

Web traffic data, Google index presence, page rank scores, and reputation-based statistical reports.

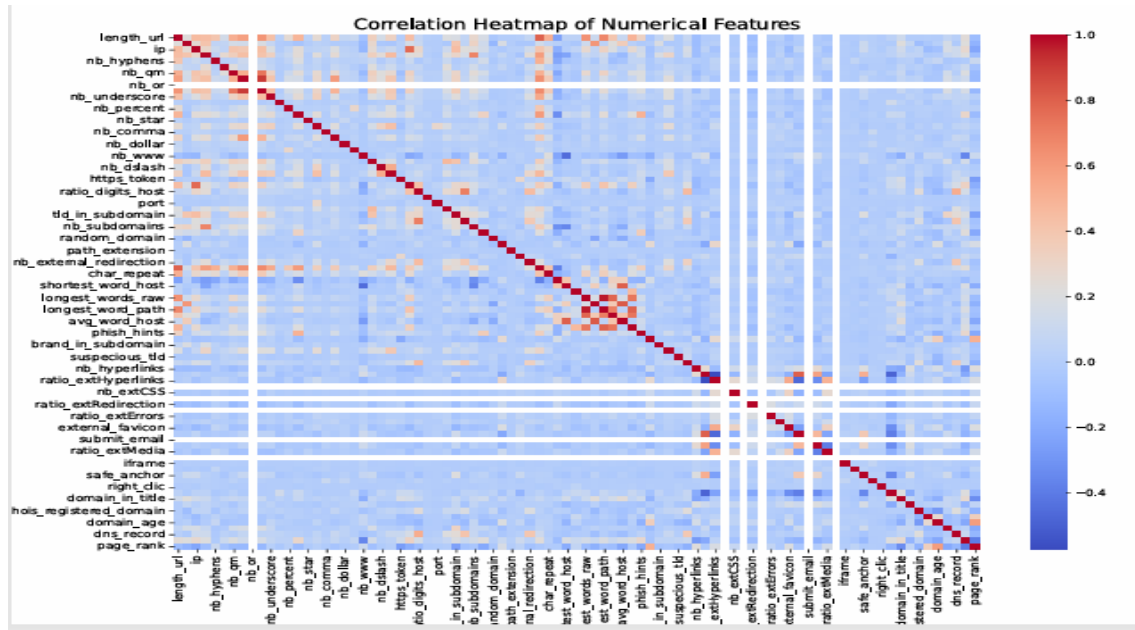
F) Target Variable

The dataset uses a binary target label: 'legitimate' or 'phishing'.

5. Insights from Dataset Exploration

- Dataset is comprehensive, combining lexical, structural, content, and reputation features.
- Most features are numeric or Boolean, making them suitable for ML models.
- Reputation-based features (traffic, indexing) are highly discriminative but may not always be available in real-time.

- Lexical and structural features are lightweight and scalable, suitable for real-time detection.
- Overall, the dataset provides a strong foundation for supervised phishing detection tasks.



Observations from the Heatmap

1. Diagonal Line (Red Band)

- The bright red diagonal indicates each feature is perfectly correlated with itself (correlation = 1).
- This is expected and confirms the heatmap is correctly showing relationships.

2. Clusters of Strong Correlation

- Certain groups of features show strong positive correlations (deep red blocks near the diagonal).
- For example, features related to **URL structure** (like length_url, length_hostname, nb_dots, nb_hyphens) are highly correlated. Longer URLs tend to have more dots and hyphens.

3. Weak or No Correlation

- Many features appear light blue, meaning they are weakly correlated or independent.
- Examples: lexical features (char_repeat, phish_hints) show little relation to domain registration details (domain_age, whois_registered_domain).

4. Negative Correlations

- Some faint blue/orange patches indicate weak negative correlations.
- Example: Higher ratio_extHyperlinks sometimes correlates with fewer internal hyperlinks, suggesting a trade-off between external vs. internal linking patterns.

5. Reputation Features Are Distinct

- Features like page_rank, web_traffic, and google_index show relatively unique patterns, not strongly tied to URL string characteristics.
- This suggests reputation-based features bring complementary information to lexical and structural features.

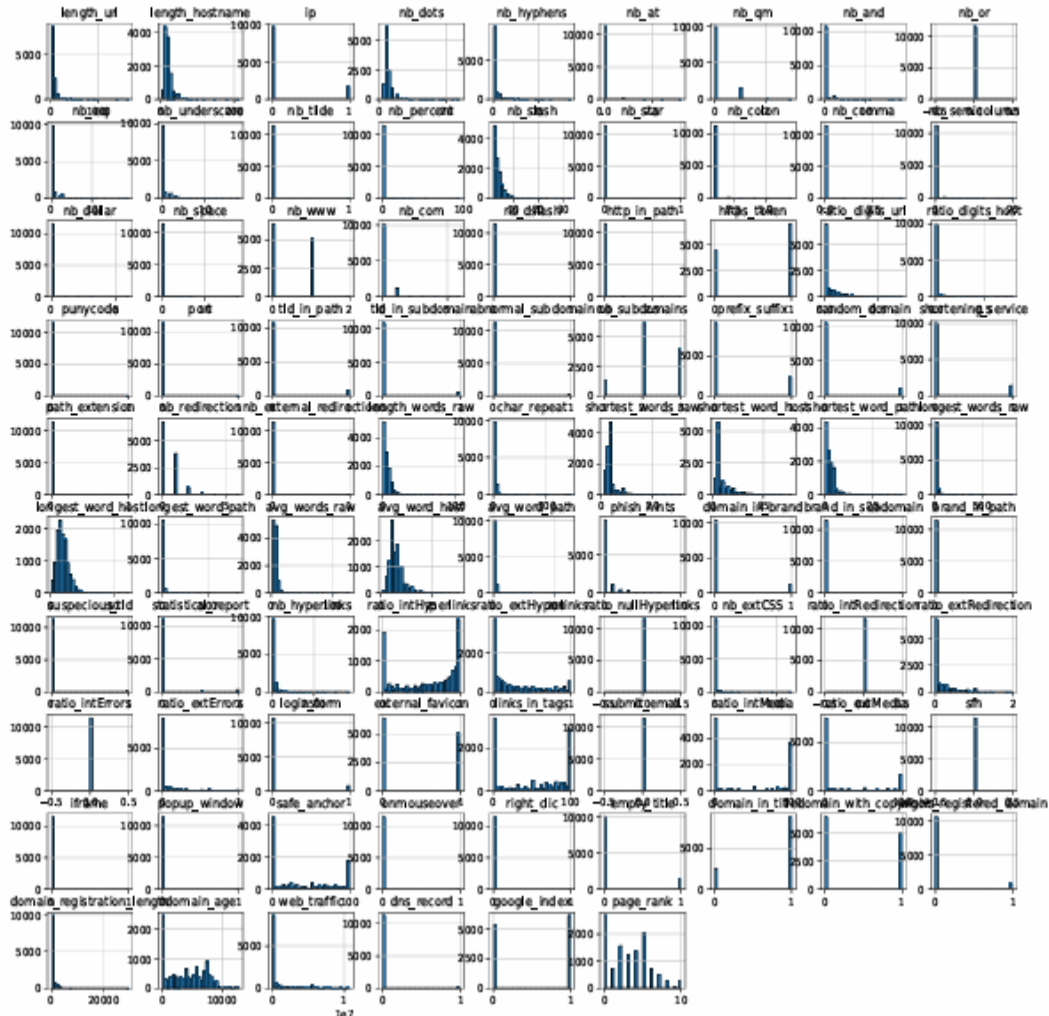
6. Implication for Modeling

- The presence of correlated groups means some features may be redundant.
- Dimensionality reduction or feature selection (like PCA or removing multicollinear variables) could improve model efficiency without losing predictive power.

Distribution of Numerical Features (Grid Plot)

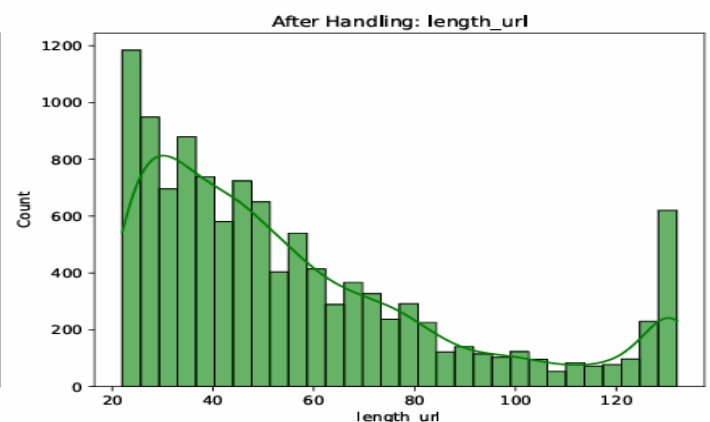
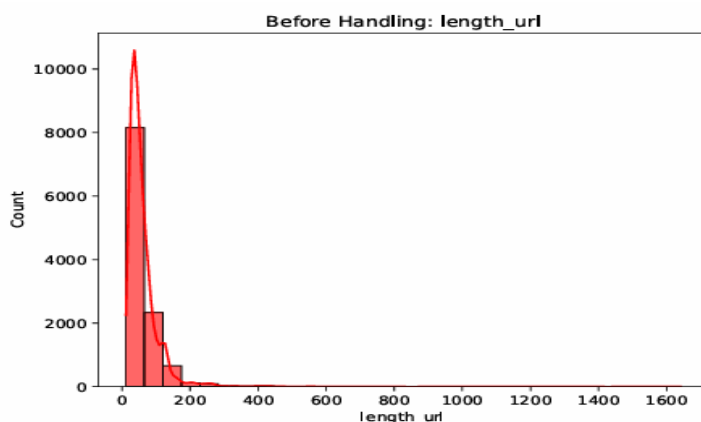
- Most features are highly skewed, with values concentrated at lower ranges.
- A few features (like page_rank, web_traffic) show broader spread.
- Skewness indicates the need for scaling or transformation before modeling.

Distribution of Numerical Features



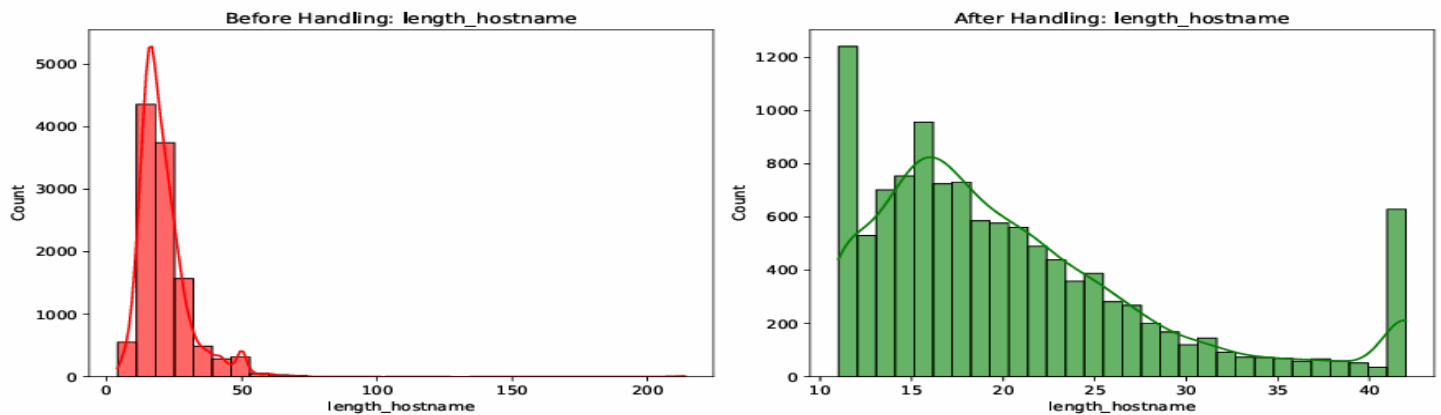
Before vs After Handling – length_url

- Before: Strong right skew, with many short URLs and few extremely long ones.
- After: Distribution appears more normalized, making the feature balanced and usable for modeling



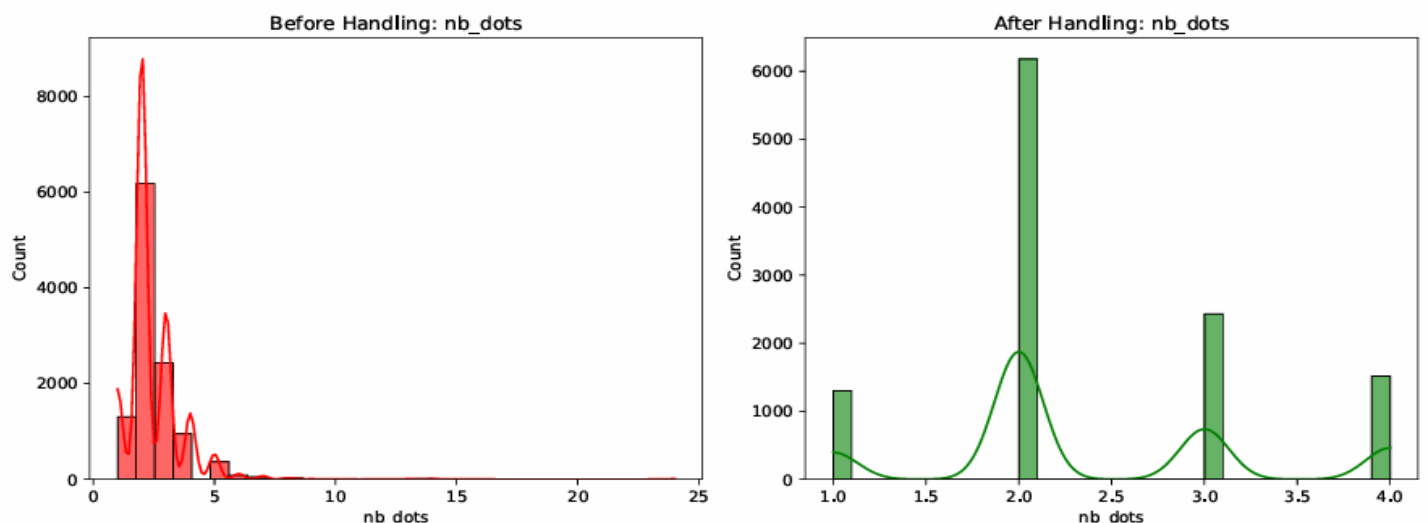
Before vs After Handling – length_hostname

- **Before:** Highly skewed towards small values with rare extreme outliers.
- **After:** More even distribution across ranges, highlighting better representation of hostnames.



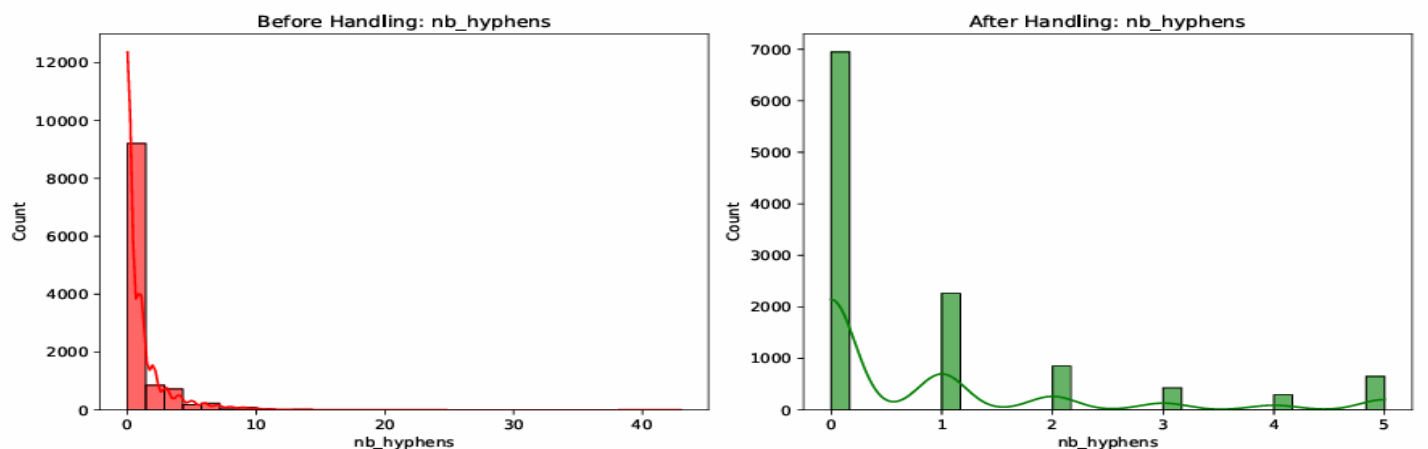
Before vs After Handling – nb_dots

- **Before:** Skewed, most URLs had very few dots, with occasional very high counts.
- **After:** Distribution flattened, showing clearer separation of normal vs. abnormal dot usage in URLs.



Before vs After Handling – nb_hyphens

- **Before:** Most URLs contained zero or very few hyphens, but some had extreme values.
- **After:** Outliers reduced; clearer focus on realistic values (0–5), aiding better pattern detection.



The findings in this report are not only theoretical but also supported with visual analysis. Each image included in the document has been described to clarify its relevance and highlight what insights it brings to phishing detection.

6. Conclusion

Detecting phishing websites is a high-stakes cybersecurity challenge that requires adaptive, data-driven solutions. Phishing detection benefits from combining URL/domain features with content-based and reputation-based analysis. Modern deep learning approaches reduce dependency on manual features, increasing adaptability. The dataset explored offers the right balance of feature diversity and real-world relevance, making it ideal for building robust, scalable, and generalizable anti-phishing models.

Code link - https://github.com/Yashaswitir28/CodeB_Internship_Phishing_Websites